QUANT

Ratio & Partnership, Mixtures & Alligations

Assignment 03

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Ratio Online Assignment -03

1. The cost of diamond varies directly as the square of its weight. Once, this diamond broke into four pieces with weights in the ratio 1:2:3: 4. When the pieces were sold, the merchant got Rs. 70,000 less. Find the original price of the diamond.

1. Rs. 1.4 lakh 2. Rs. 2 lakh 3.Rs. 1 lakh 4.Rs. 2.1 lakh

2. A student gets an aggregate of 60% marks in five subjects in the ratio 10 : 9 : 8 : 7 : 6. If the passing marks are 50% of the maximum marks and each subject has the same maximum marks, in how many subjects did he pass the examination?

1. 2 2. 3 3.4 4.5

3. The ratio of sum of squares of first n natural numbers to square of sum of first n natural numbers is 17:325. The value of n is

1. 15 2. 25 3.35 4.30

4. Two men, sitting on the dining table. One man has 7 breads and other had 5 breads. A third man passing by requested them to share their food in return for money. The three of them shared the breads equally and the third traveller paid the other two a total of Rs 24. Find the difference between the amounts received by first two men?

1. Rs. 18 2. Rs. 12 3.Rs. 16 4.Rs. 15

5. Points A and B are both in the line segment PQ and on the same side of its midpoint. A divides PQ in the ratio 2:3, and B divides PQ in the ratio 3:4. If AB = 2, then the length of PQ is

1. 70 2. 75 3.80 4.85

6. On a certain day, the ratio of the passenger in the Business class and the Economy class travelling by Plane is 1:3. The ratio of the fares collected from each Business class and Economy class passengers is 30:1. If the total amount collected from all the passengers is Rs 1,320. Find the amount collected from the Economy class passengers.

1. Rs 240 2. Rs 360 3.Rs 480 4.Rs 120

7. By mistake, instead of dividing Rs 564 among three persons P, Q and R in the ratio 1/3:1/4:1/5, it was divided in the ratio 3:4:5. Who gains the most and how much?

1. P, Rs. 99 2. Q, Rs. 30 3.R, Rs. 72 4.R, Rs. 91

8. 20% of the students in a class participated in sports. Out of the students who participated, 75% were males. Male students who participated in sports constitute 90% of the economically poor students in the class. What is the ratio of the number of economically poor students to the number of students in the class?

1. 1:4 2.1:5 3.5:6 4.1:6

9. The ratio of incomes of company A and company B is 3:5 and the ratio of their expenditures is 2:3. Who does save more? (You have to assume that their incomes are more than their respective expenditure.)

1. Company A
2. Company B
3. Both save equally
4. Depends upon the incomes of A and B

10. In a house, there are dogs, cats and parrot in the ratio 3:7:5. If the number of cats was more than the number of dogs by a multiple of both 9 and 7, what is the minimum of pets in the house?

1. 945
2. 630
3. 252
4. 238

Answer key & Explanations

Q No.	Key	Explanations
1.	3	Let the original weight of the diamond be $10x$. Hence, its original price will be $k(100x^2)$ . . . where k is a constant. The weights of the pieces after breaking are x, $2x$, $3x$ and $4x$. Therefore, their prices will be kx^2, $4kx^2$, $9kx^2$ and $16kx^2$. So the total price of the pieces = $(1 + 4 + 9 + 16)\ kx^2 = 30kx^2$. Hence, the difference in the price of the original diamond and its pieces = $100kx^2 - 30kx^2 = 70kx^2 = 70000$. Hence, $kx^2 = 1000$ and the original price = $100kx^2 = 100 \times 1000 = 100000$ = Rs. 1 lakh.
2.	3	Let his marks be 100, 90, 80, 70 and 60 in the five subjects. Hence, totally he has scored 400 marks. This constitutes only 60% of the total marks. Hence, total marks = $\frac{400}{0.6} = 667$, which incidentally is the maximum marks in all 5 subjects. Since the total marks in each subject is the same, hence maximum marks in each subject will be $\left(\frac{667}{5}\right) \approx 133$. Out of this 50% is the passing marks. In other words, to pass in a subject he needs to score 66.5 marks. We can see that only in one subject he scored less than this, viz 60. Hence, he passed in 4 subjects.
3.	2	Sum of squares of first n natural numbers $= \frac{n(n+1)(2n+1)}{6}$ Squares of sum of first n natural numbers $= \left[\frac{n(n+1)}{2}\right]^2$ Now the ratio is17:325 So $\frac{\frac{n(n+1)(2n+1)}{6}}{\left[\frac{n(n+1)}{2}\right]^2} = \frac{17}{325}$ Thus, by solving we get n= **25**.
4.	2	As the two men had a total of 12 breads. The first and the second man must have given 3 breads and 1 bread to the third man. Hence, the ratio of their share must be 3:1. Now third person paid Rs. 24 for 4 breads. So cost of one bread is Rs 6. Hence the first and the second man get Rs.18 and Rs. 6 respectively. The 1st man gets Rs. 12 more than the second.
5.	1	Let PA=2x and AB=3x and PB=3y and BQ=4y PB:BQ=3:4, PA:QA=2:3, PQ=5x=7y $x=\frac{7}{5}y$ ---------- (i) Now, AB=PQ−PA−BQ=7y−4y−2x => 3y−2x=2 ---------- (ii) From equation (i) and (ii) y=10 and Hence, PQ = **70.**
6.	4	Let the number of passengers travelling by Business class and Economy class be x and 3x respectively. Lets the fares collected from each of the Business class and Economy class passengers be 30y and y respectively. Hence x(30y)+3x(y)=30xy+3xy=33xy=1320 xy =40. Total amount collected from the Economy class =3xy=3×40= Rs 120
7.	4	Ratio 1/3:1/4:1/5 is equivalent to 20:15:12. So in this case, P, Q and R would have got Rs 240, Rs 180 and Rs 144 respectively. But actually the money was divided in the ratio 3:4:5

		and shares of P, Q and R in this case was 141, 188 and 235 respectively. So R got Rs. 91 more than the actual value.
8.	4	Let the total number of students in the class be 't' Let total number of students in the class who are economically poor be 'p' Total number of students who participated =t/5 Total number of male students who participated $=\frac{3}{4}\times\frac{t}{5}=\frac{9p}{10}\Rightarrow\frac{p}{t}=\frac{1}{6}$ => So required ratio = 1: 6.
9.	2	The ratio of income of A and B is 3:5. Ratio of their expenditures is 2:3 i.e. 3:4.5.since ratio of their expenditures is 3:4.5 only. Obviously savings of B will be something more than savings of A.
10.	1	Let the number of dogs, cats and parrot be 3k,7k,5k.Then 7k – 3k = 63p So k=63p/4.Min. Value of p for which k is natural number is 4 thus k = 63. So No. of pets = 15k = 15 × 63 = 945.